Cognitive Symmetry: Modeling Human Regressive Saccades via LLM Attention Entropy and Surprisal

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Abstract

This research proposal outlines a comparative study between human eye-tracking behavior and the internal processing states of Large Language Models (LLMs). We specifically investigate "look-backs" or regressive saccades in human readers ,typically occurring at a 30% rate during syntactic disambiguation and correlate them with LLM **Surprisal** and **Attention Entropy**. By utilizing the Dundee and GECO corpora, we aim to identify if specific transformer layers function as "syntax-checkers" that mirror human cognitive re-processing.

1 Introduction

Human language processing is non-linear. When readers encounter "garden-path" sentences (e.g., *"The old man the boat"*), they experience a breakdown in their initial parse, leading to regressive saccades (looking back at previous words). Previous work by Futrell et al. (2019) suggests LLMs can act as "psycholinguistic subjects." This study extends that framework by analyzing whether LLM uncertainty ,quantified through Information Theory metrics ,aligns with physical eye-movement data.

2 Proposed Methodology

We propose a three-axis evaluation involving multiple model architectures and high-granularity eye-tracking datasets.

2.1 Metrics of Uncertainty

- **LLM Surprisal (S):** Calculated as $S(w_i) = -\log P(w_i|w_{<i})$. We hypothesize that high surprisal at the disambiguating word triggers human regressions and LLM entropy spikes.
- **Attention Entropy (H):** For an attention head, $H(A_i) = -\sum_j A_{ij} \log A_{ij}$. A spike in H indicates the model is "searching" the context, mirroring the human "look-back" mechanism.
- **Regression-In/Out Rate:** Extracted from human corpora to measure the probability of a reader returning to a token.

3 Experimental Setup

3.1 Data Selection and Acquisition

We utilize three primary corpora that provide word-level gaze data and explicit regression markers:

1. **GECO (Ghent Eye-Tracking Corpus):** Provides *Regression-Path Duration* (Go-Past time) for a full novel [Cop et al., 2017].
2. **ZuCo 2.0:** Provides simultaneous eye-tracking and EEG data [Hollenstein et al., 2019].
3. **Dundee Eye-Tracking Corpus:** Containing 51,000 tokens of editorial text, it is the most robust source for *Regression-In* metrics [Kennedy and Pynte, 2003]. **Due to the unique syntactic complexity and unavailability of this corpus, we strongly request that this dataset be made available to us.**

3.2 Large Language Models (LLMs)

To test the scaling hypothesis, we will evaluate:

- **Baseline:** GPT-2 (Small) and BERT-Base.
- **Mid-Tier:** Llama 3 (8B) and Mistral-7B.
- **High-Tier:** Llama 3 (70B) to observe if human-like "cognitive alignment" emerges with scale.

4 Novel Contributions

4.1 The Regression-Entropy Lag

We will analyze the temporal alignment. While humans physically look back *after* the disambiguating word, does the LLM show high entropy *at* the word or across the subsequent context window?

4.2 Layer-wise Functional Localization

We will identify "Saccadic Heads" ,specific attention heads that consistently broaden their focus (high entropy) only when surprisal is high. Identifying these provides a functional interpretability map of the transformer architecture.

5 Expected Results

We expect a statistically significant correlation ($r > 0.4$) between Attention Entropy and human regression rates. Furthermore, we expect that larger models (Llama 3 70B) will demonstrate a "tighter" alignment with human scanpaths than smaller models, suggesting that superior language modeling requires human-like processing strategies.

6 Conclusion

By bridging the gap between physical eye movements and abstract attention weights, this study provides a new lens into LLM interpretability and the computational nature of human syntax processing.

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